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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (Even)**	**Course Name**	**PE-III: Lab -Reinforcement Learning**
Semester	**6th**	**Course Code**	**23ADS1644**
Roll No	**13**	**Name of Student**	**Prerna Utane**

Practical Number	**04**
Course Outcome	CO1: Apply fundamental reinforcement learning concepts to model and solve decision-making problems. CO2: Implement basic reinforcement learning algorithms to train agents in a given environment. CO3: Analyze agent behavior
Aim	To apply K-Means clustering to group data points into clusters based on similarity with and without labeled output.
Problem Definition	The objective of this experiment is to apply the K-Means clustering algorithm to group data points into clusters based on similarity. Using the California housing dataset, clustering is performed on geographical features such as longitude and latitude to identify natural groupings of locations. The aim is to analyze how data points can be grouped without labeled outputs (unsupervised learning) and understand the spatial distribution of clusters using visualization techniques.
Theory (100 words)	K-Means clustering is an unsupervised machine learning algorithm used to partition data into K distinct clusters based on similarity. It works by initializing K centroids, assigning each data point to the nearest centroid, and updating centroids iteratively until convergence. The objective is to minimize the within-cluster sum of squares (WCSS), ensuring that points in the same cluster are as similar as possible. Unlike supervised learning, K-Means does not require labeled data. The algorithm is sensitive to the choice of K and initial centroids. It is widely used for pattern recognition, data segmentation, and exploratory data analysis. Visualization helps in understanding how clusters are formed and how well-separated they are.

Procedure and Execution (100 Words)	Algorithm: 1. Import required libraries (Pandas, NumPy, Matplotlib, Scikit-learn). 2. Load the dataset using Pandas. 3. Select relevant features (longitude and latitude). 4. Store feature values in a variable (X_housing). 5. Choose the number of clusters (K). 6. Initialize the K-Means model with chosen K. 7. Fit the model to the data. 8. Predict cluster labels for each data point. 9. Extract cluster centers (centroids). 10. Visualize clusters using a scatter plot. 11. Plot centroids to show cluster centers.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

df_housing = pd.read_csv('/content/sample_data/california_housing_train.csv')

print("First 5 rows of the dataset:")
display(df_housing.head())
print("\nDataFrame Info:")
df_housing.info()

feature_columns = ['longitude', 'latitude']
X_housing = df_housing[feature_columns].values

print(f"\nShape of features for clustering (X_housing): {X_housing.shape}")

kmeans_housing = KMeans(n_clusters=n_clusters_housing, random_state=42,
n_init=10)
kmeans_housing.fit(X_housing)
y_kmeans_pred_housing = kmeans_housing.labels_

print("K-Means predicted cluster assignments for the first 10 data points:",
y_kmeans_pred_housing[:10])
```

```
plt.figure(figsize=(10, 8))
plt.scatter(X_housing[:, 0], X_housing[:, 1], c=y_kmeans_pred_housing, s=10,
cmap='viridis', alpha=0.6)
centers_housing = kmeans_housing.cluster_centers_
plt.scatter(centers_housing[:, 0], centers_housing[:, 1], c='red', s=200, alpha=0.9,
marker='X', label='Cluster Centers')
plt.title(f'K-Means Clustering Result on California Housing
(K={n_clusters_housing})')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.colorbar(label='Predicted Cluster')
plt.legend()
plt.grid(True)
plt.show()
```

Output:

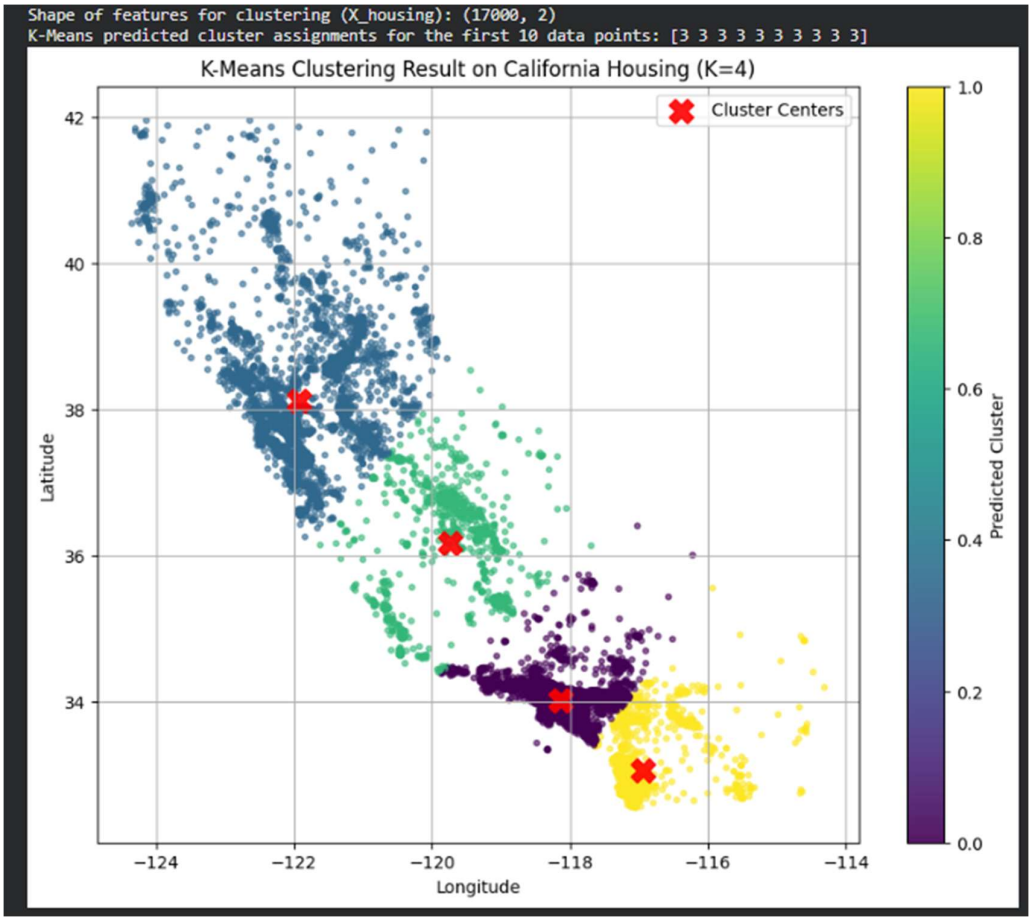
1. **Dataset**

```
First 5 rows of the dataset:
```

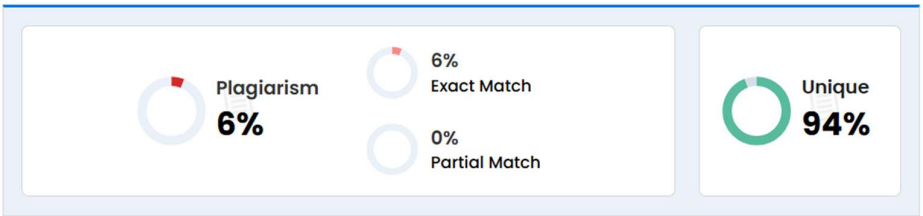
	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	472.0	1.4936	66900.0
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	463.0	1.8200	80100.0
2	-114.56	33.69	17.0	720.0	174.0	333.0	117.0	1.6509	85700.0
3	-114.57	33.64	14.0	1501.0	337.0	515.0	226.0	3.1917	73400.0
4	-114.57	33.57	20.0	1454.0	326.0	624.0	262.0	1.9250	65500.0

```
DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17000 entries, 0 to 16999
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           17000 non-null  float64
 1   latitude            17000 non-null  float64
 2   housing_median_age  17000 non-null  float64
 3   total_rooms         17000 non-null  float64
 4   total_bedrooms      17000 non-null  float64
 5   population          17000 non-null  float64
 6   households          17000 non-null  float64
 7   median_income       17000 non-null  float64
 8   median_house_value  17000 non-null  float64
dtypes: float64(9)
memory usage: 1.2 MB
```

2. K-mean Cluster Graph



Output Analysis	• The scatter plot displays clusters formed based on geographical coordinates (longitude and latitude). • Each color represents a different cluster assigned by the K-Means algorithm. • The red 'X' markers indicate the cluster centroids, which are the central points of each cluster. • The clustering shows how locations are grouped spatially, revealing patterns such as densely populated or geographically similar regions. • Since no labeled output is used, clustering is purely based on feature similarity.

	• The choice of K significantly affects the results; different values of K may produce different cluster structures. • Overlapping clusters may indicate that features used are not sufficient to clearly separate data.
Link of student Github profile where lab assignment has been uploaded	https://github.com/23071121-pixel/RL_Practicals/edit/main/Practical-4_13
Conclusion	The K-Means clustering algorithm effectively groups data points based on similarity without requiring labeled data. In this experiment, geographical features were used to form clusters, demonstrating spatial grouping in the dataset. While the model provides useful insights into data distribution, its performance depends on the choice of K and selected features. Overall, K-Means is a simple yet powerful tool for exploratory data analysis and pattern discovery.
Plag Report (Similarity index < 12%)	
Date	28/03/2026